

D-RAG: Implementation Roadmap

Konstantinos Kritsotakis – Vasilis Seitaridis

1. Objective

The goal of this project is to implement a simplified version of the **D-RAG (Differentiable Retrieval-Augmented Generation)** framework for Knowledge Graph Question Answering (KGQA). Our implementation will preserve the main contribution of the paper: **differentiable fact retrieval through Gumbel-Softmax sampling and end-to-end optimization of the retriever and generator**.

Since the original implementation requires very large language models and high-end GPU resources, we will build a lightweight version that reproduces the core algorithmic ideas while remaining computationally feasible.

2. Implementation Plan

The implementation will be completed in four stages.

Stage 1 – Repository Analysis and Environment Setup

- Study the available D-RAG implementation.
- Install all required dependencies.
- Execute the retriever module independently.
- Understand the data pipeline and repository structure.

Stage 2 – Retriever Implementation

Implement and validate the differentiable retriever, including:

- Knowledge graph representation
- Fact embeddings
- Graph Neural Network (GNN)
- Fact scoring
- Gumbel-Softmax differentiable sampling

The objective of this stage is to obtain differentiable fact selection probabilities.

Stage 3 – Lightweight End-to-End D-RAG

Instead of the original large language model, we will implement a lightweight neural generator while preserving the original training pipeline.

The resulting system will follow:

Question



Retriever



Fact probabilities



Gumbel-Softmax



Selected facts



Neural Generator



Answer

This allows us to reproduce the core differentiable optimization proposed by the paper.

Stage 4 – Experimental Evaluation

We will evaluate the implementation through several experiments.

Planned experiments include:

- Hard Top-k Retrieval vs Differentiable Gumbel-Softmax Retrieval
- Different Gumbel-Softmax temperature values
- Robustness to noisy facts
- Retrieval quality analysis

Performance will be evaluated using prediction accuracy together with retrieval behavior.

3. Planned Extension

Instead of only reproducing the original work, we plan to extend the implementation by comparing classical discrete retrieval with differentiable retrieval.

This comparison will allow us to investigate the practical impact of differentiable retrieval on the final question answering performance.

Expected Outcome

By the end of the project, we expect to obtain a working prototype of D-RAG that demonstrates the main contribution of the paper, namely differentiable retrieval and end-to-end optimization, together with experimental comparisons against classical retrieval methods.